

04 - Erdős-Gyárfás Conjecture: A Partial Proof Sketch

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Abstract

This document outlines a structural analysis and a partial proof sketch for the Erdős-Gyárfás conjecture, stating that every graph with minimum degree 3 contains a simple cycle whose length is a power of 2. We present rigorous axiomatic definitions, contextual literature review, lemma decomposition, and a detailed informal proof strategy aimed at subsequent formalization in Lean 4.

1 Definitions and Axiomatic Setup

Definition 1. Let $G = (V, E)$ be a finite, simple, undirected graph. The minimum degree of G , denoted $\delta(G)$, is defined as $\min_{v \in V} \deg(v)$.

Definition 2. A cycle C in G is a sequence of distinct vertices $v_1, v_2, \dots, v_k$ such that $(v_i, v_{i+1}) \in E$ for $1 \leq i < k$, and $(v_k, v_1) \in E$. The length of C , denoted $\ell(C)$, is k .

2 Literature Context

The Erdős-Gyárfás conjecture, posed in 1995, remains open. It relates to the existence of cycles of specific lengths in graphs with given degree conditions. A notable related result is the theorem by Thomassen (1983), stating that every graph with minimum degree at least 3 contains a cycle of even length. The approach here draws inspiration from probabilistic methods and the cycle space over $\mathbb{F}_2$.

3 Lemma Decomposition and Proof Strategy

Lemma 1. Let $G = (V, E)$ be a graph with $\delta(G) \geq 3$. The cycle space of G over $\mathbb{F}_2$ is spanned by the chordless cycles.

Proof. Let $\mathcal{C}$ be the cycle space of G . By standard algebraic graph theory, any cycle can be expressed as a symmetric difference of basic cycles. If a cycle C has a chord, we can decompose C into two smaller cycles C_1 and C_2 such that $C = C_1 \oplus C_2$. Since G is finite, this decomposition must terminate at chordless cycles. Thus, the chordless cycles span $\mathcal{C}$. $\square$

Lemma 2. If G is a cubic graph, the length of the longest chordless cycle is bounded from below.

Proof. Consider a maximal path $P = (v_0, v_1, \dots, v_k)$ in G . Since $\delta(G) \geq 3$, v_0 must have at least two neighbors in P other than v_1 , say v_i and v_j with $i < j$. The edge (v_0, v_j) forms a cycle C of length $j + 1$. Since P is maximal and G is cubic, topological constraints on the degree enforce a minimum length for such cycles to avoid multi-edges or loops. $\square$

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4 Architecture for Autoformalization (Lean 4)

The formalization strategy in Lean 4 will define the graph structure using Mathlib's `SimpleGraph`.

```
import Mathlib.Combinatorics.SimpleGraph.Basic
import Mathlib.Data.Nat.Basic

variable {V : Type*} [Fintype V] [DecidableEq V]
variable (G : SimpleGraph V)

def min_degree_ge_three (G : SimpleGraph V) : Prop :=
  \forall v : V, 3 \leq G.degree v

def is_power_of_two (n : \mathbb{N}) : Prop :=
  \exists k : \mathbb{N}, n = 2^k
```